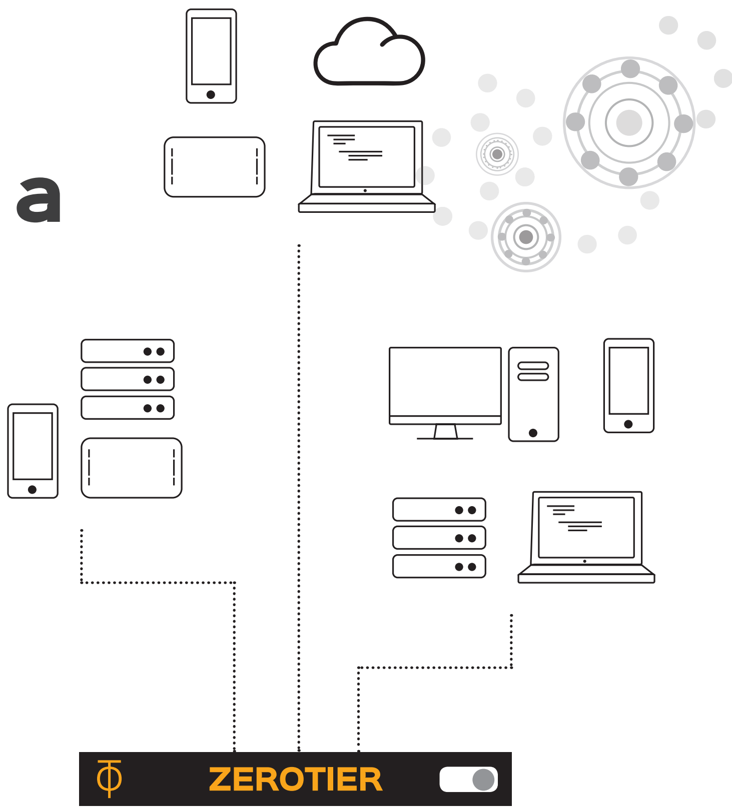
ZeroTier:
The Internet as a Global Switch



ZeroTier describes itself as a global Ethernet switch to connect any number of devices. The subscribers connect to this imaginary switch and then communicate as if they were in the same local network. Let's look into how this actually works.

ZeroTier's Ethernet switch is figurative and symbolic; subscribers actually connect via a virtual private network (VPN) to a predefined VPN server operated by ZeroTier. The VPN server connects clients' devices so that they can talk directly to each other. The result is an encrypted peer-to-peer network.

The transport network is responsible for securing participants' communication and authentication. This network also achieves another important task by finding the best path between clients. Above this network layer, ZeroTier emulates the well-proven Ethernet to provide applications with a known protocol. For the participants, the VPN appears as an Ethernet switch. The technology behind this design is similar to IPsec and VXLAN.

The data packets between ZeroTier clients are encrypted and only understood by other clients, but not by the ZeroTier servers. The used algorithm originates from the toolbox of elliptic curves and offers a key length of 256 bits. The client software automatically handles the key exchange and certificate management.

In Figure 1, the clients use the ZeroTier cloud as a fully meshed backbone network that independently finds the shortest connections between all sites.

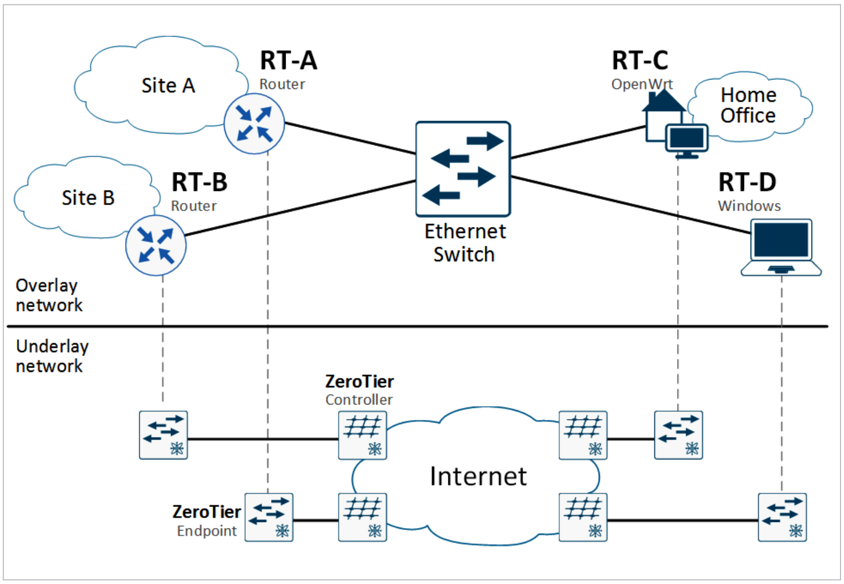


Figure 1: Various clients connect through the ZeroTier switch